

SAKSHI MAURYA

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SUMMARY

Software engineer with 2+ years at Intel and Viasat, including 1 year 5 months full time as an SDE, and an M.Tech in CS from NIT Delhi (CGPA 9.11). Builds systems software in C++ and automation platforms in Python on Linux: epoll based concurrent servers, raw socket packet analysis, Xeon platform validation and CI/CD test infrastructure. Debugs across application, protocol and firmware layers. First author on a paper accepted to a Springer LNNS conference.

TECHNICAL SKILLS

- **Languages:** C, C++, Python, Go, SQL
- **Systems and Networking:** Linux, Socket Programming, epoll, Multithreading, Concurrency, Non-blocking I/O, Raw Sockets, libpcap, TCP/IP, UDP, ICMP, 4G/LTE RAN, GTP Tunneling
- **Testing and CI/CD:** pytest, Test Automation Framework Design, Regression Suites, Root Cause Analysis, Jira, Zephyr, Jenkins, GitHub Actions, Git, Docker, Kubernetes, Bash
- **Cloud, Data and ML:** AWS (EC2, S3), Apache Kafka, Grafana, Prometheus, Real-Time Telemetry, scikit-learn, NumPy, Pandas, Anomaly Detection, LangChain, RAG, Chroma, LLM API Integration

WORK EXPERIENCE

Intel — System Software Engineering Graduate Intern, Bengaluru Jun 2025 – Apr 2026

- Automated the full build, deploy and run validation flow for next generation Xeon server platforms with 35+ Python test cases, cutting engineer effort from roughly 28 hours per cycle to 8 and letting runs complete largely unattended.
- Built a 4 node Xeon monitoring platform streaming real time hardware telemetry through Apache Kafka, applying machine learning anomaly detection to flag abnormal node behavior on Grafana dashboards.
- Developed an LLM and RAG log triage assistant using LangChain with a Chroma vector store over system logs and runbooks, returning probable root cause and fix recommendations. Debugged failures on pre release Linux server platforms and reported defects to platform and firmware teams.

Viasat — Software Development Engineer 1, Chennai Jun 2022 – Oct 2023

- Designed and built 45+ Python automation scripts validating the 4G/LTE RAN access node, hardware accelerator and authentication modules, replacing fully manual verification across 3 network domains.
- Owned integration of the suites into Jenkins for continuous build validation, moving protocol regression detection from the release stage into the development cycle, and root caused protocol layer failures to the responsible layer with development teams.

EDUCATION

National Institute of Technology, Delhi | M.Tech, Computer Science and Engineering 2024 – 2026
CGPA: 9.11/10

Lovely Professional University | B.Tech, Computer Science and Engineering 2018 – 2022
CGPA: 8.03/10

TECHNICAL PROJECTS

Full Stack Task Manager with CI/CD (Python, React, Kubernetes) | GitHub Jun 2026 – Aug 2026

- Containerized a React, Flask and MongoDB app with Docker, deployed to Kubernetes on AWS EC2 via a GitHub Actions build/test/deploy pipeline gated by pytest.

Network Packet Sniffer and Protocol Analyzer (C++) | GitHub Apr 2026 – May 2026

- Built a packet capture application in C++ using raw sockets and libpcap, parsing Ethernet, TCP, UDP and ICMP headers in real time, with per flow state tracking across 5 million packets.

High-Performance Multi-threaded TCP Server (C++) | GitHub Feb 2026 – Mar 2026

- Built a concurrent TCP server in C++ on an epoll event loop with a worker thread pool, non blocking I/O and connection pooling, architected for 5,000+ concurrent connections; managed connection lifecycle and request queueing explicitly to bound memory use and avoid thread exhaustion under sustained load.

PUBLICATION

S. Maurya, A. Naz and K. Verma, “Debris-Aware Network Management for 6G Non-Terrestrial Networks: An Application Function Approach,” 2nd Int. Conf. on Next-Generation Networks and Deployable AI (NGNDAI 2026), Springer LNNS. *Accepted; to appear.*

CERTIFICATIONS

Machine Learning in Production (MLOps) | Building and Evaluating Advanced RAG | LangChain for LLM Application Development | ChatGPT Prompt Engineering for Developers | Deep Learning Specialization | Machine Learning Specialization